

Server & Active Directory Troubleshooting Guide

Windows Server, AD DS, DNS, and file services field guide

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Server rule #1: verify last-known-good backup exists BEFORE risky operations (BVT-IRP-051). Rule #2: on domain controllers, never 'just restore a snapshot' of a DC in a multi-DC domain — USN rollback will poison replication; use proper AD restore methods.

Active Directory / Authentication

Symptom	Diagnosis & Fix
Users can't log on (domain-wide)	Check DC health: 'dcdiag /v', time skew >5 min breaks Kerberos ('w32tm /query /status'), DNS on clients MUST point only to DCs. Fix time hierarchy (PDC syncs external, rest sync domain).
Trust relationship failed (one PC)	Machine account password mismatch — 'Test-ComputerSecureChannel -Repair' or rejoin; investigate if many machines hit at once (snapshot rollback?)
Account lockouts repeating	Find source: lockout events 4740 on PDC emulator show caller computer; usual suspects = stale creds in mapped drives, scheduled tasks, mobile mail profiles, disconnected RDP sessions
GPO not applying	'gpresult /h report.html' — check security filtering, WMI filters, slow-link detection, and SYSVOL replication health ('dfsrdiag' / event 2213 for DFSR)
Replication errors	'repadmin /replsummary' + '/showrepl'; resolve DNS first (most replication failures are DNS); check for tombstone-lifetime exceedance before forcing

DNS / DHCP

- DNS: scavenging misconfigured causes stale records or vanishing records — set aging deliberately; forwarders healthy? ('Resolve-DnsName' against each); never point a DC's own DNS at external resolvers (loopback/partner DC only).
- DHCP: scope exhaustion (shorten lease for guest-heavy networks), rogue DHCP detection (dual offers), failover partnership status if configured.

File Services

Symptom	Diagnosis & Fix
'Access denied' despite correct group	Kerberos token stale (log off/on), share vs NTFS mismatch (effective access tool), nested-group over 1,015 SIDs token bloat
Files locked 'in use'	Computer Management → Open Files → close handle; identify the app holding it; educate on proper close

Symptom	Diagnosis & Fix
Slow file access	SMB signing/encryption CPU cost, antivirus scanning storms on the share, network path (see BVT-RUN-058), disk latency (PerfMon: Avg Disk sec/Transfer >25ms = storage problem)
Previous Versions empty	Verify Shadow Copies schedule + storage allocation; not a substitute for backup

General Server Health

- Disk space: alert at 15%, act at 10% — cleanup order: temp files, old logs/IIS logs, superseded updates ('Dism /StartComponentCleanup'), THEN grow disk. Never let C: hit zero on a DC or Exchange box.
- Unexpected reboots: Event 41/6008 → check for automatic-update reboot policy, PSU/power, memory (run memtest during window).
- Services failing to start: check dependent-service chain and service-account password expiry (common after credential rotations — coordinate with BVT-POL-032 rotations).
- Certificate expirations: inventory with expiry alerts at 30 days — expired certs masquerade as 'app broken' tickets.